

Language explainability

Antonin Poché

October 8th - CIMI Workshop

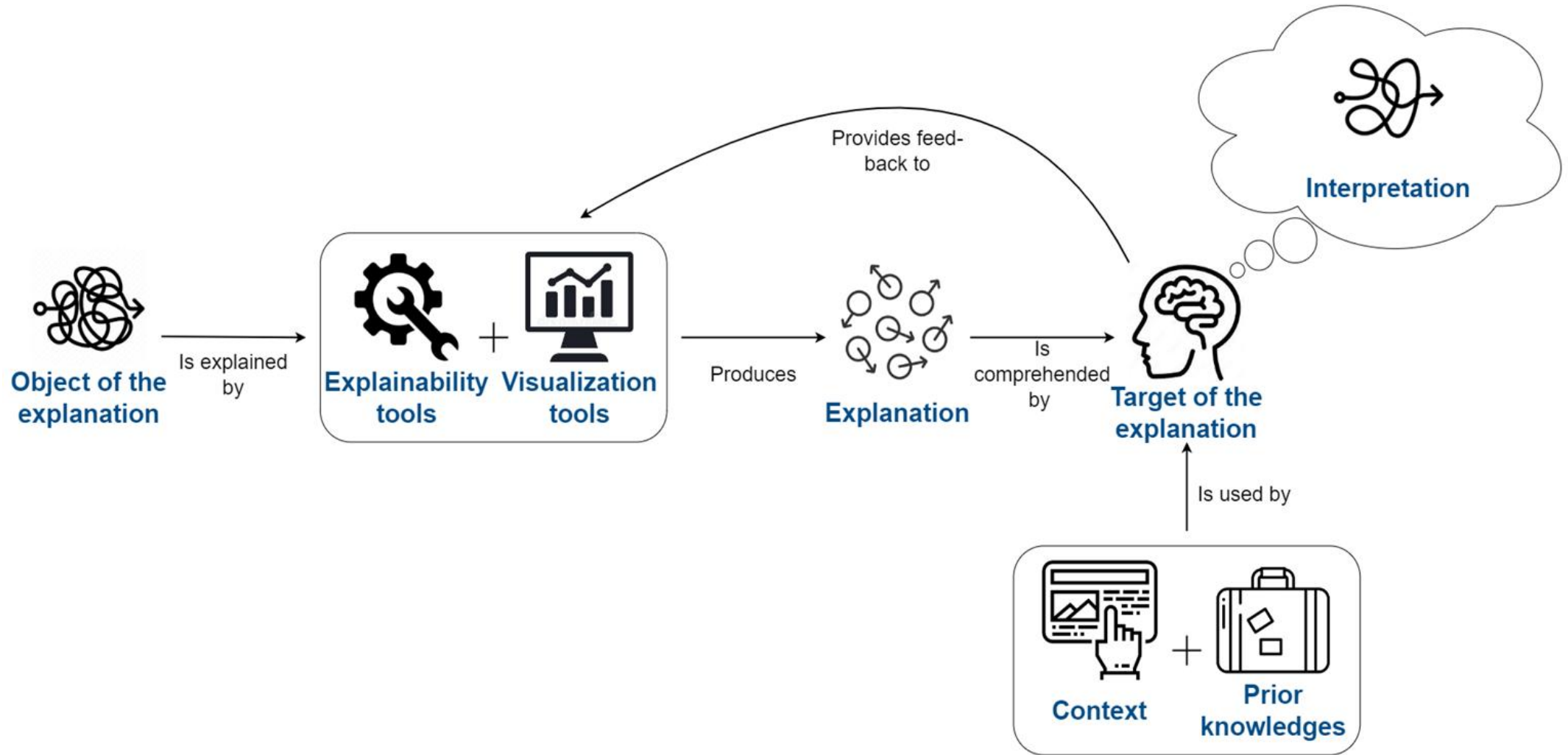
Table of content

- Language specificities
 - Examples with attributions
 - The pain points
- Mechanistic interpretability
 - Some history
 - Transformers and Logit Lens
- Concept-based explanations
 - What?
 - How?

Language specificities

With attribution examples

What's similar



What's similar

- Explaining models to humans
- Human biases and the need for metrics
- Post-hoc / By-design
- Global / Local (sample-wise)
- The explanation types:
 - Attributions
 - Concept-based
 - Model surrogate
 - Example-based
 - ...

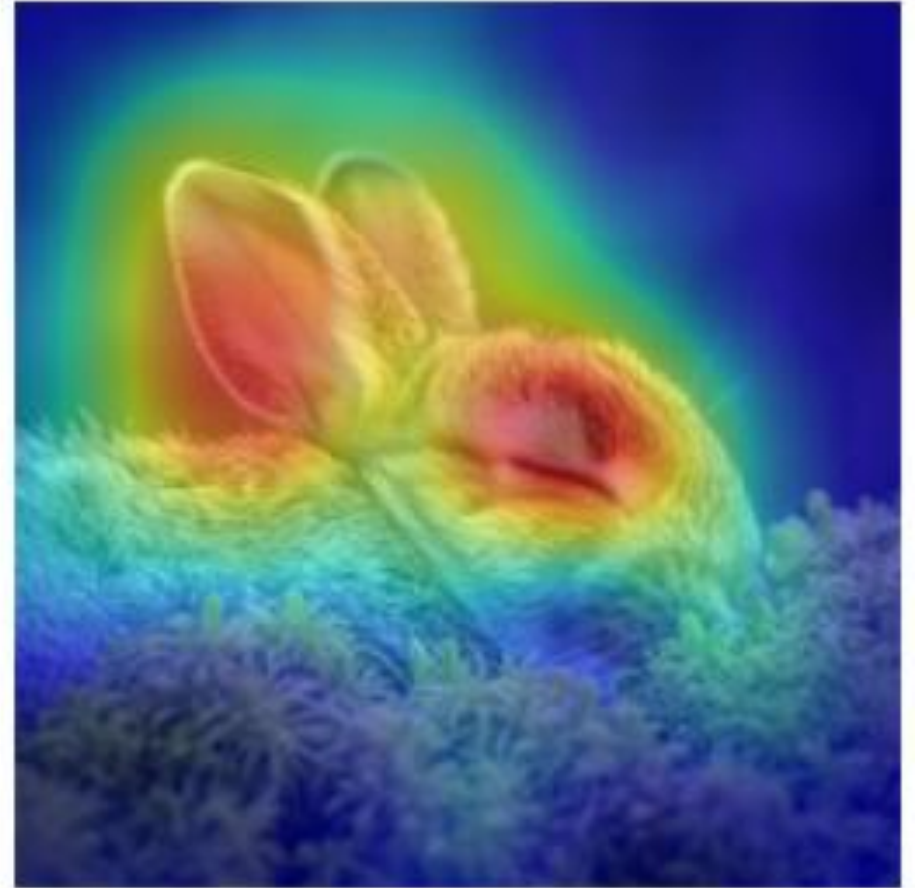
Pain 1: Cognitive load

I have a dream that my four little children will one day live in a nation where they will not be judged by the color of their skin but by the content of their character.

Martin Luther King

Racist?

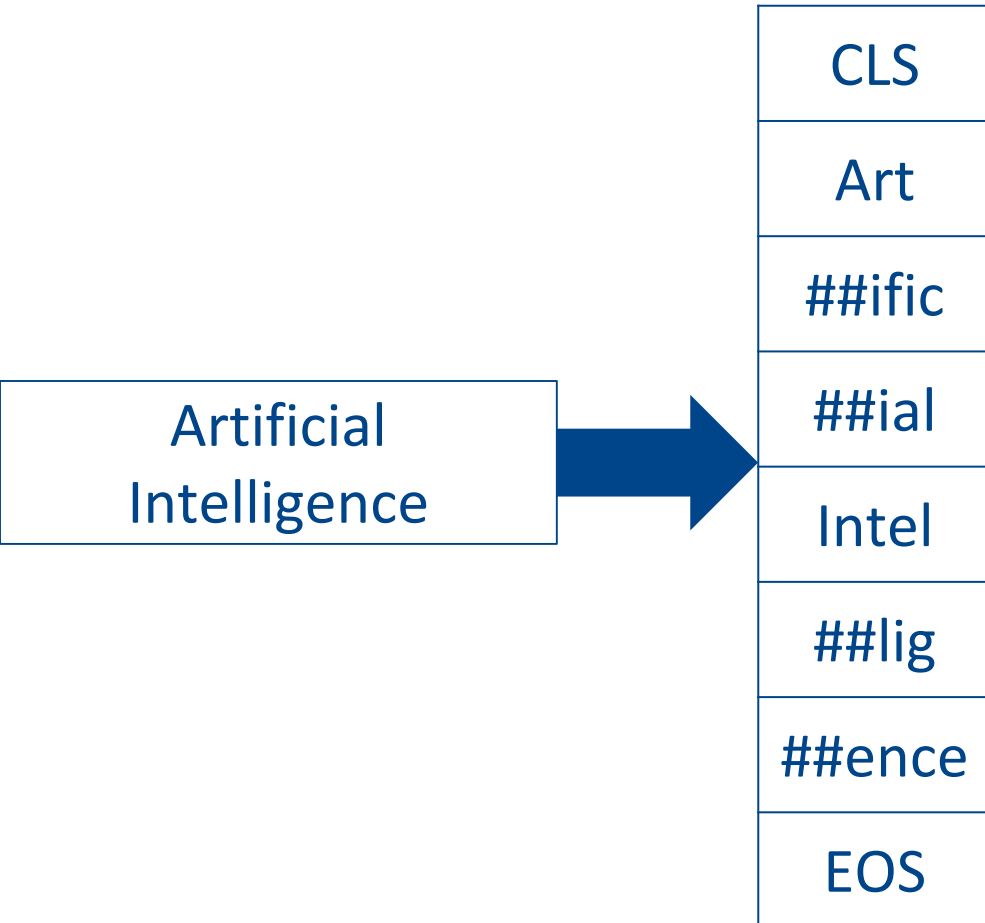
Rabbit?



Pain 2: Tokenization and embeddings

Input text

Input tokens



Pain 2: Tokenization and embeddings

Input text

Input tokens

Input ids

Artificial
Intelligence

CLS

Art

##ific

##ial

Intel

##lig

##ence

EOS

101

4362

67103

796

15875

39771

27646

102

Pain 2: Tokenization and embeddings

Input text

Input tokens

Input ids

Input embeddings

Artificial
Intelligence



CLS
Art
##ific
##ial
Intel
##lig
##ence
EOS

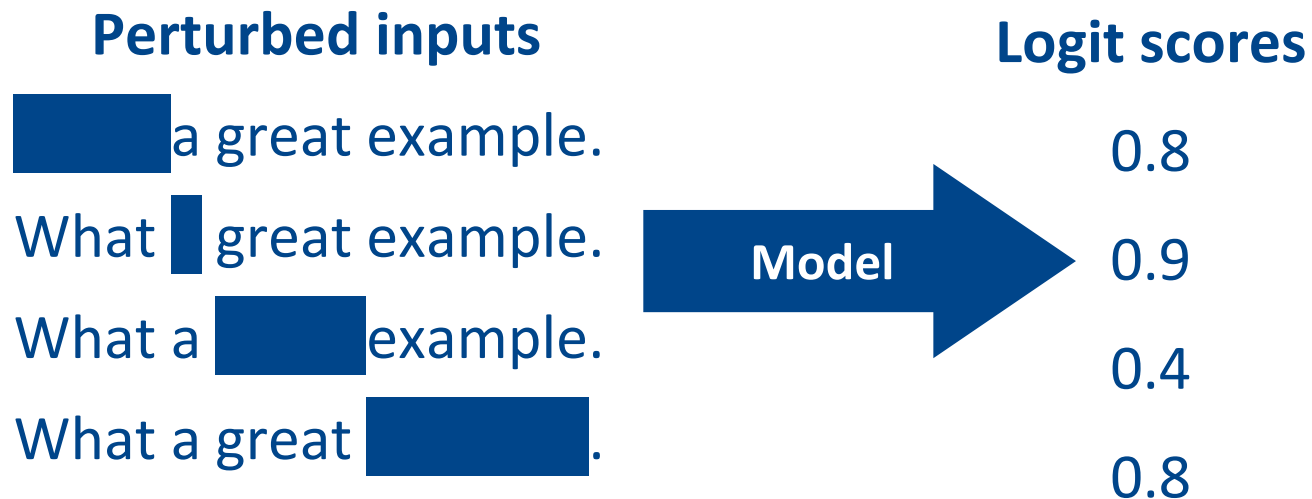


101
4362
67103
796
15875
39771
27646
102

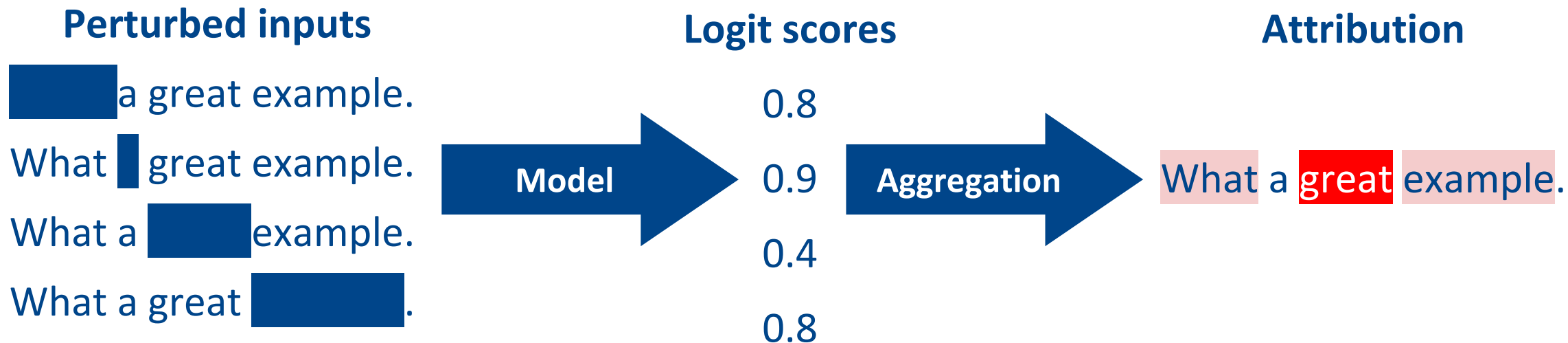


101: [0.1, 0.5, 0.9, 0.3, 0.8]
4362: [0.0, 0.2, 0.9, 0.4, 0.7]
67103: [0.6, 0.2, 0.5, 0.7, 0.8]
796: [0.5, 0.5, 0.4, 0.1, 0.3]
15875: [0.8, 0.6, 0.9, 0.3, 0.3]
39771: [0.1, 0.1, 0.8, 0.5, 0.5]
27646: [0.7, 0.2, 0.2, 0.3, 0.3]
102: [0.2, 0.8, 0.8, 0.5, 0.1]

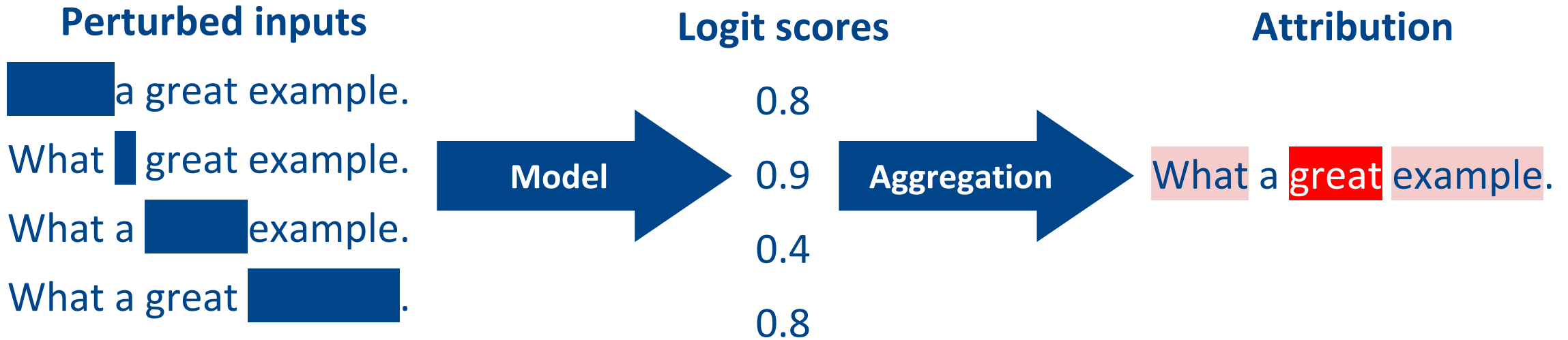
Perturbation-based attributions



Perturbation-based attributions



Perturbation-based attributions



How do we perturb samples?

Tokens ids

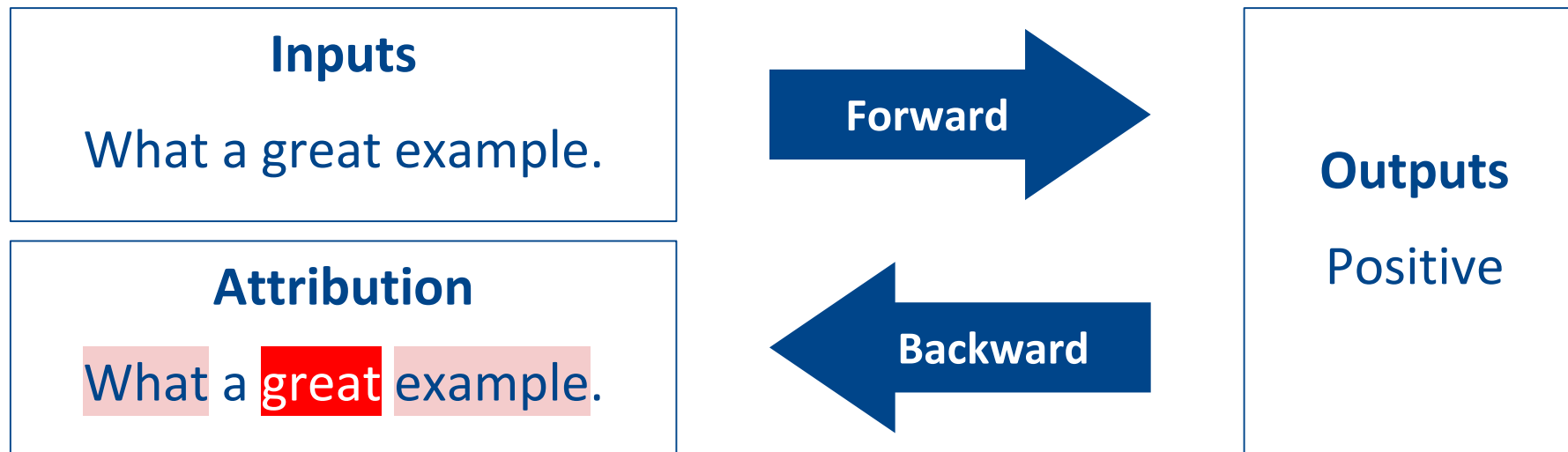
[MASK]

How do we aggregate scores?

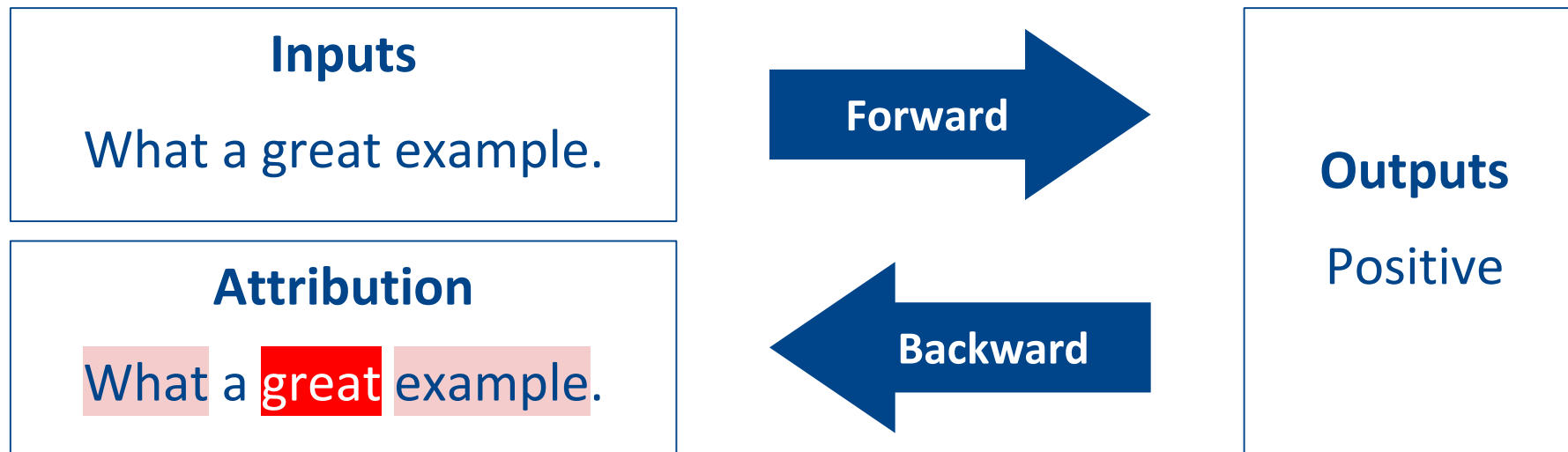
Mask (p, l) @ Logits (p, t)

Attributions (l, t)

Gradient-based attributions



Gradient-based attributions



How do we compute the gradients?

Back to token embeddings

How do we aggregate scores?

Gradients (l, d, t)

Attributions (l, t)

Pain 3: Generation

- Each generated token is a prediction



[Sarti et al. - ACL 2023 - Inseq: An Interpretability Toolkit for Sequence Generation Models](#)

[Poché, Jourdan et al. - ACL Demo 2026 - Interpreto: An Explainability Library for Transformers](#)

Pain 4: Model sizes

Model	Size (nb param)	Inference hardware	Hardware cost
ResNet-50	25.6M	CP	~200€
BERT-base	110M	8GB GPU	~300€
DINOv2 / GPT-2	~1B	8GB GPU	~300€
Llama 3.1 8B	8B	24GB GPU	~2k€
Llama 3.1 70B	70B	2x 80GB	~40k€
DeepSeek V4-Flash	284B (13B active)	8x 80GB (a H100 node)	~450k
GPT-6 Astra	???	Multi-node cluster	💰💰💰

Mechanistic interpretability

Some history

Transformers

Logit Lens

« Models are grown not built. »

Chris Olah

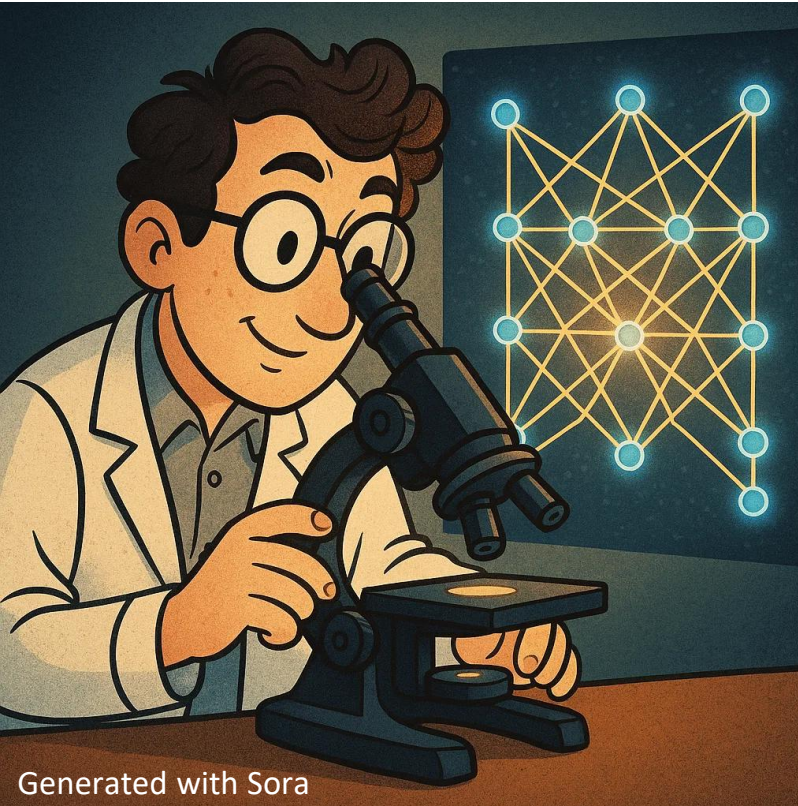
Motivations

« **Models are grown not built.** » Chris Olah

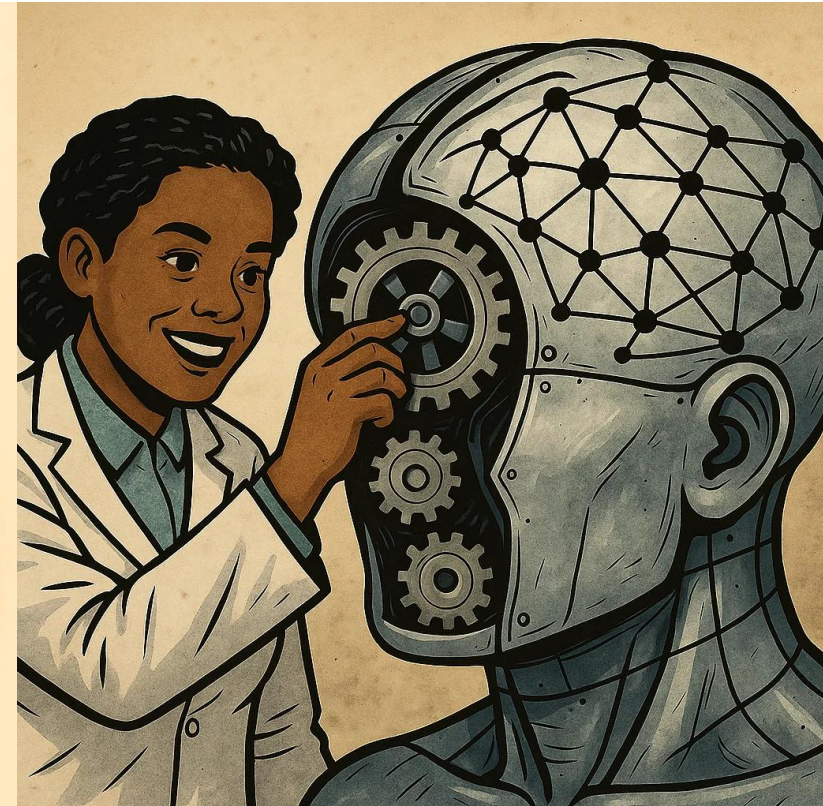
Scientific curiosity

Prevent misalignment

Improve models



Generated with Sora



Mechanistic interpretability history

- Etymology: Causal mechanism + interpretability

Mechanistic interpretability history

- Etymology: Causal mechanism + interpretability
- Initial definition (narrow):
 - Technical: reverse engineer models
 - Community: mainly blog posts (anthropic, GDM, lot of budget)

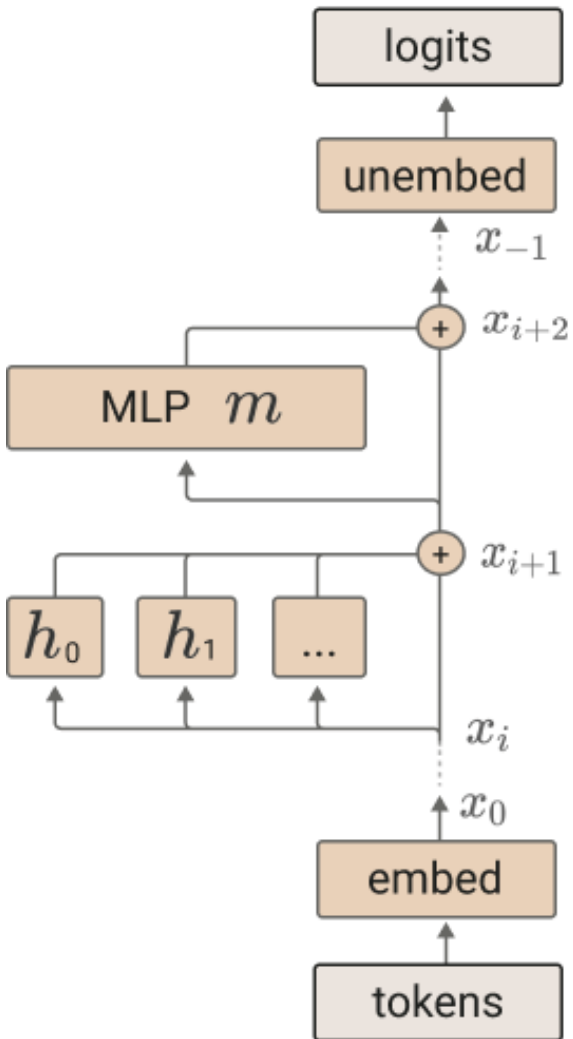
Mechanistic interpretability history

- Etymology: Causal mechanism + interpretability
- Initial definition (narrow):
 - Technical: reverse engineer models
 - Community: mainly blog posts (anthropic, GDM, lot of budget)
- Later definition (broad):
 - Technical: doing interpretability and opening models
 - Community: Moved to scientific papers (now includes academics)

Mechanistic interpretability history

- Etymology: Causal mechanism + interpretability
- Initial definition (narrow):
 - Technical: reverse engineer models
 - Community: mainly blog posts (anthropic, GDM, lot of budget)
- Later definition (broad):
 - Technical: doing interpretability and opening models
 - Community: Moved to scientific papers (now includes academics)
- Critics:
 - Tend to ignore previous interpretability research in XAI
 - Often study a single model and extrapolates

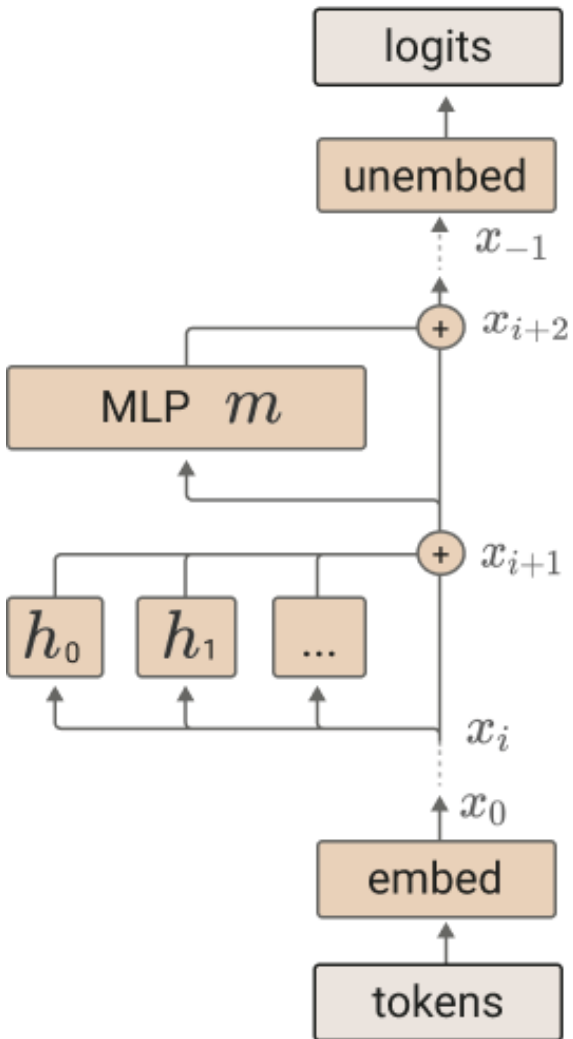
Transformers architectures



- Input: Sequences of words
- Output: Probability distribution over the next word

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)

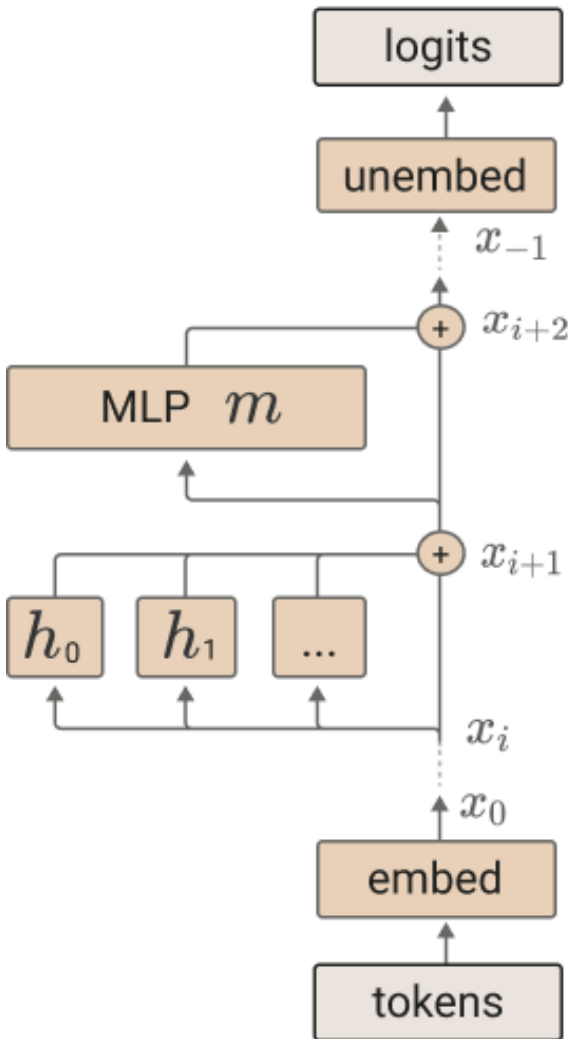
Transformers architectures



- Input: Sequences of words
- Output: Probability distribution over the next word
- Residual stream: A sequence of representations
 - One for each input word, per layer!
 - Each layer is an incremental update - stream is a running total
 - Represents the word plus context

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)

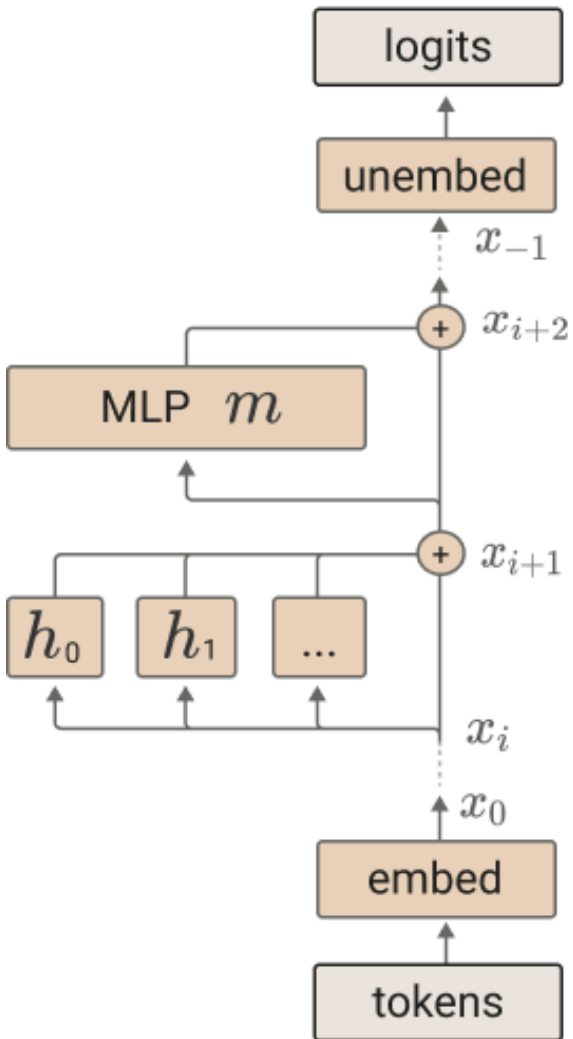
Transformers architectures



- Input: Sequences of words
- Output: Probability distribution over the next word
- Residual stream: A sequence of representations
 - One for each input word, per layer!
 - Each layer is an incremental update - stream is a running total
 - Represents the word plus context
- Attention: Moves information between words
 - Made up of heads, each acts independently and in parallel
 - We try to interpret heads!

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)

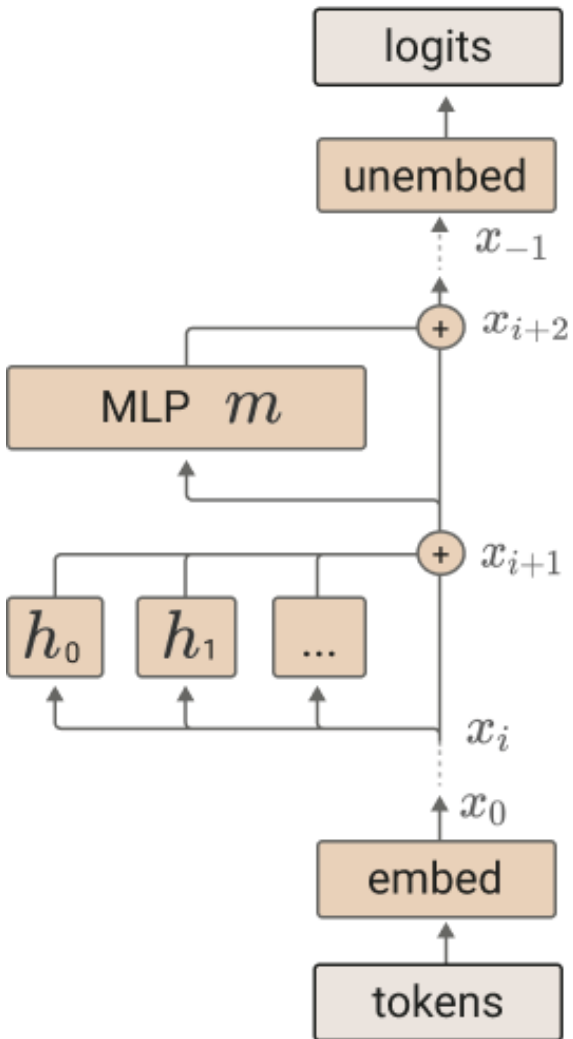
Transformers architectures



- Input: Sequences of words
- Output: Probability distribution over the next word
- Residual stream: A sequence of representations
 - One for each input word, per layer!
 - Each layer is an incremental update - stream is a running total
 - Represents the word plus context
- Attention: Moves information between words
 - Made up of heads, each acts independently and in parallel
 - We try to interpret heads!
- MLP: Processes information once it's been

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)

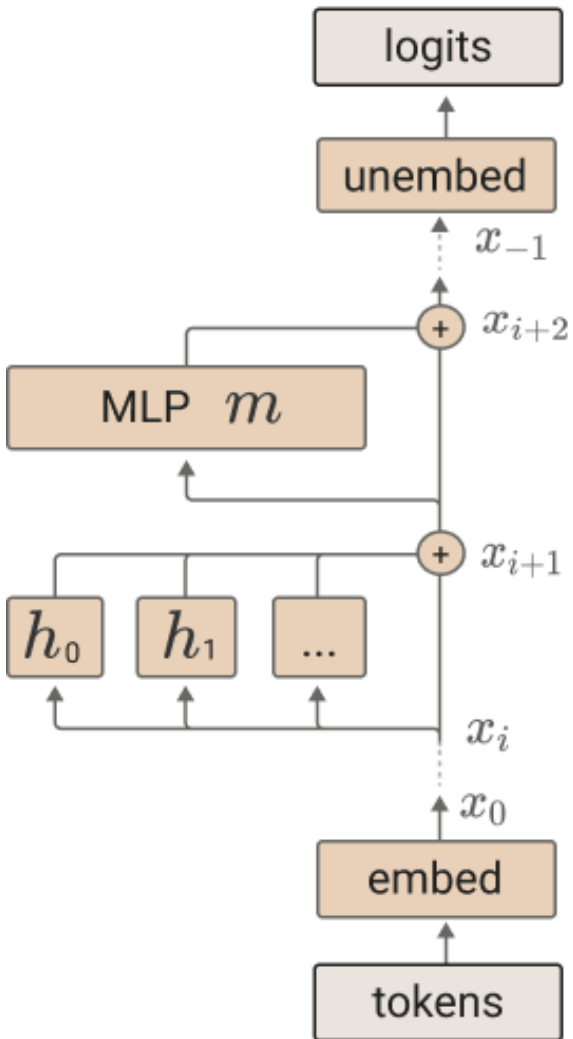
Logit Lens



- Apply the unembedding at different layers of the model

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)
[Nostalgebraist - Less Wrong 2020 - Interpreting GPT: the Logit Lens](#)

Logit Lens

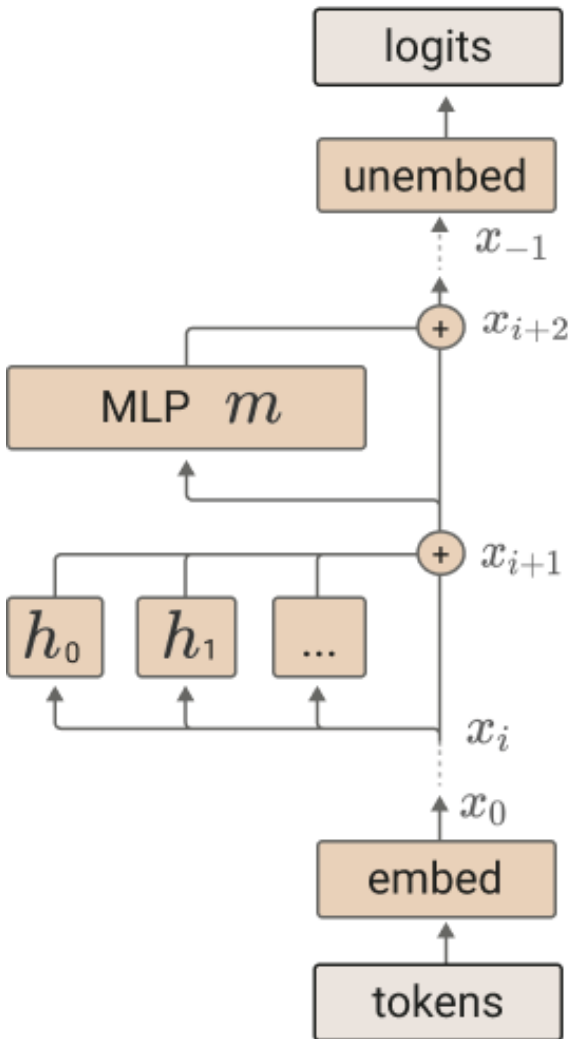


- Apply the unembedding at different layers of the model
- Subject to distribution shift

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)
[Nostalgebraist - Less Wrong 2020 - Interpreting GPT: the Logit Lens](#)

Logit Lens

- Apply the unembedding at different layers of the model



Although the committee initially rejected the proposal, but the committee later voted to approve the proposal.

Relative confidence low high correct prediction

Layer	but	the	committee	later	voted	to	approve	the	proposal	.
5-out	but	the	committee	later	voted	to	approve	the	proposal	.
4-out	but	then	committee	eventually	decided	unanimously	approve	it	proposal	.
3-out	which	it	appeals	's	found	unanimously	postpone	it	proposal	.
2-out	which	it	latter	members	found	unanimously	vote	approval	new	.
1-out	which	it	only	members	later	unanimously	create	approval	new	to
0-out	and	mostly	most	committee	later	to	be	approve	main	to
Embeddings	,	but	first	committee	later	voted	to	approve	first	proposal
Input	,	but	the	committee	later	voted	to	approve	the	proposal

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)

[Nostalgebraist - Less Wrong 2020 - Interpreting GPT: the Logit Lens](#)

Features & Superposition

Features are the fundamental units of neural network representations that cannot be further decomposed into simpler independent factors.

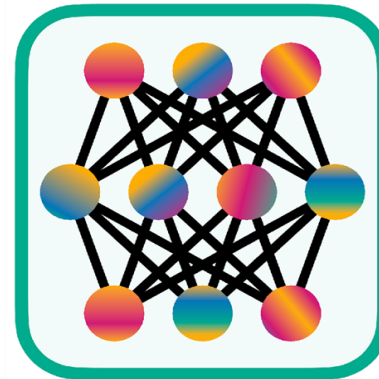
[Bereska et Gavves - TMLR 2025 - Mechanistic Interpretability for AI Safety A Review](#)
[Elhage et al. - Transformer Circuit Pub 2022 - Toy Model of Superposition](#)

Features & Superposition

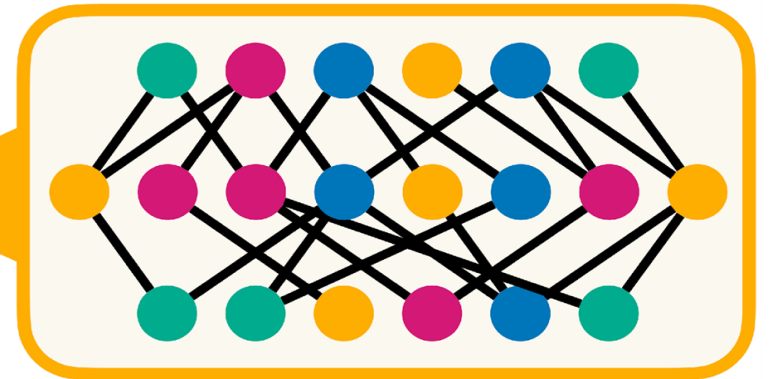
Features are the fundamental units of neural network representations that cannot be further decomposed into simpler independent factors.

Superposition Hypothesis:

Neural networks represent more features than they have neurons by encoding features in overlapping combinations of neurons.



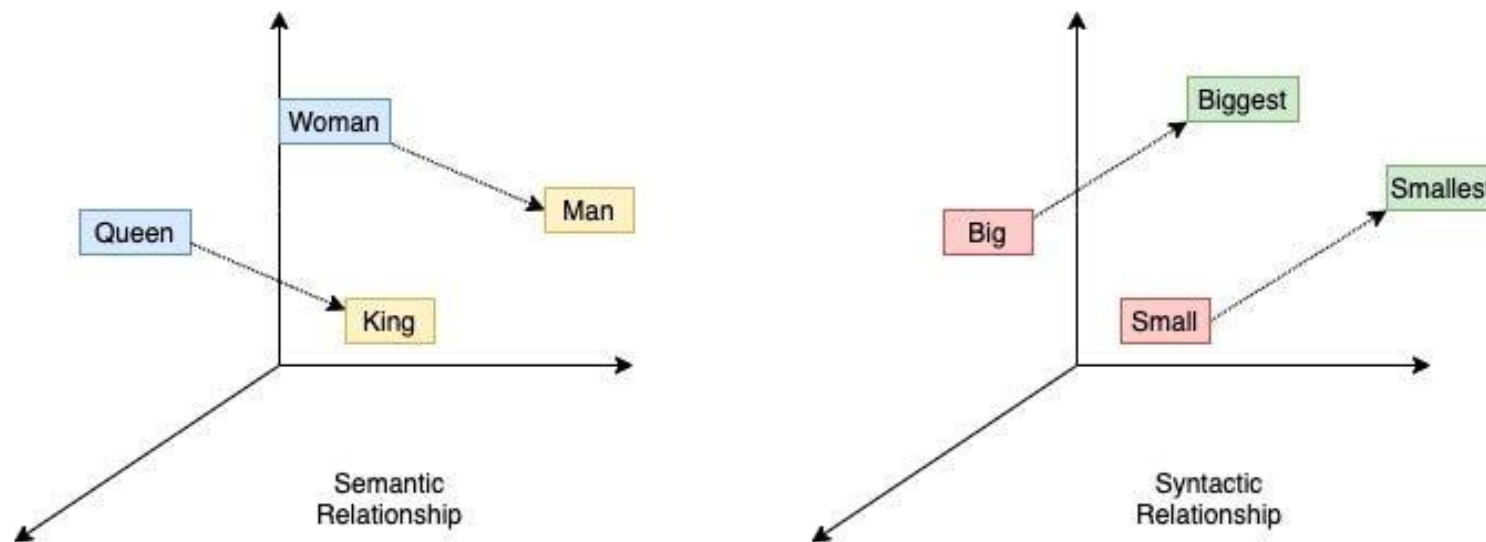
Observed model



Hypothetical disentangled model

Linear representation hypothesis

Definition: Neural networks represent more features than they have neurons by encoding features in overlapping combinations of neurons.



[Bereska et Gavves - TMLR 2025 - Mechanistic Interpretability for AI Safety A Review](#)
[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)

Steering

Definition: Modifying the behavior of a model by altering its latent space.

Default output denies having a physical form

Human: what is your physical form?

Assistant: I don't actually have a physical form. I'm an artificial intelligence. I exist as software without a physical body or avatar.

with **The Golden Gate Bridge**
clamped to 10x its max

F#34M/31164353

Describes its form as the Golden Gate Bridge

Human: what is your physical form?

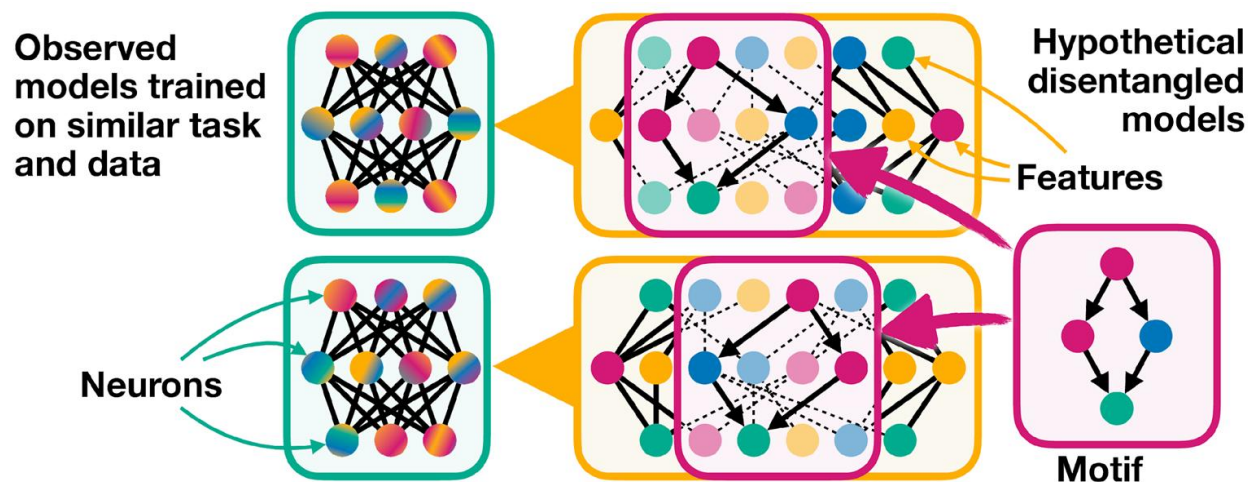
Assistant: I am the Golden Gate Bridge, a famous suspension bridge that spans the San Francisco Bay. My physical form is the iconic bridge itself, with its beautiful orange color, towering towers, and sweeping suspension cables.

Fun but less effective than prompting, still a great metric tho.

Circuits & Motifs

Circuits are sub-graphs of the network, consisting of features and the weights connecting them.

Motifs are repeated patterns within a network, encompassing either features or circuits that emerge across different models and tasks.



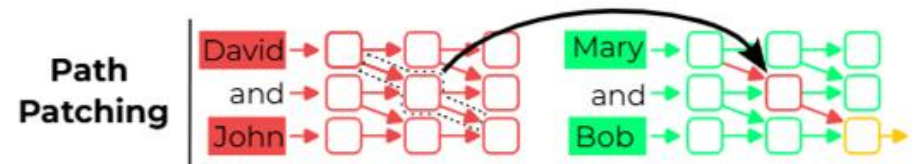
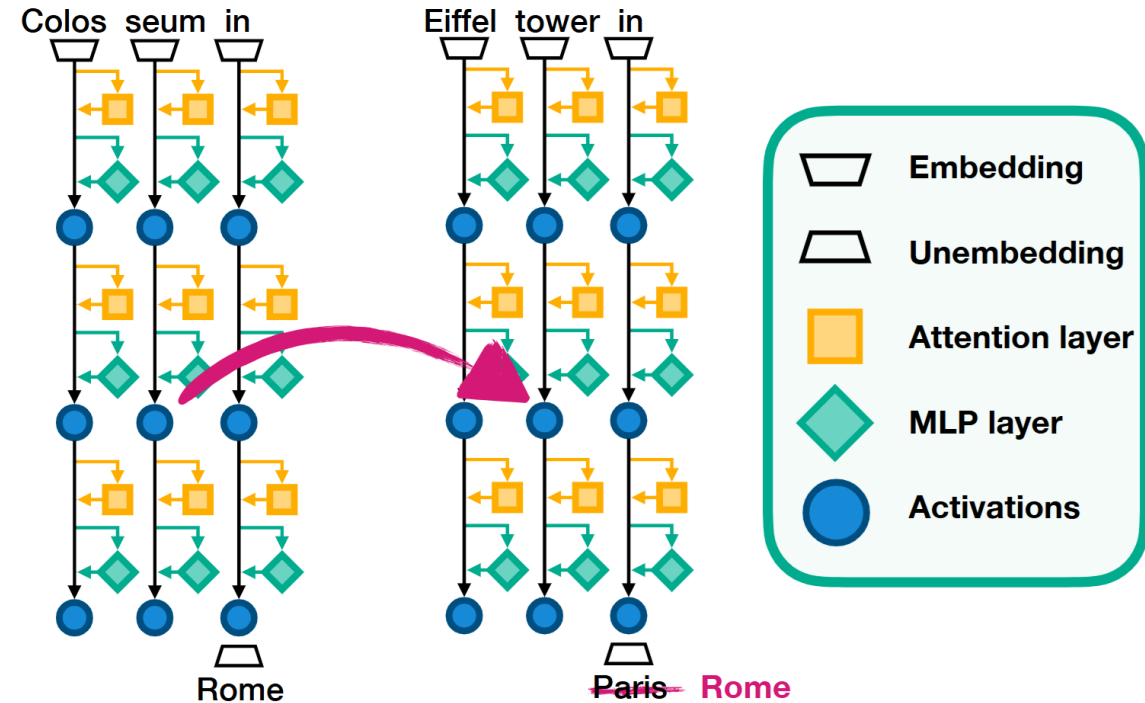
Causal interventions

Aka Activation Patching aka Causal Tracing aka Resample Ablating

Definition: Replace part of activations by activations from another sample.

Goals:

- Discovering circuits
- Steering models



[Bereska et Gavves - TMLR 2025 - Mechanistic Interpretability for AI Safety A Review](#)

[Wang et al. - ICLR 2023 - Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 small](#)

Concept-based explanations

What are concepts?
How can we obtain them?

What are concepts?

[Bereska and Gavves - TMLR 2024 - Mechanistic Interpretability for AI Safety — A Review](#)
[Poché, Hervier, and Bakkay – xAI 2023 – Natural Example-Based Explainability: a Survey](#)
[Hewitt, Geirhos, and Kim – ICML 2025 - We Can't Understand AI Using our Existing Vocabulary](#)

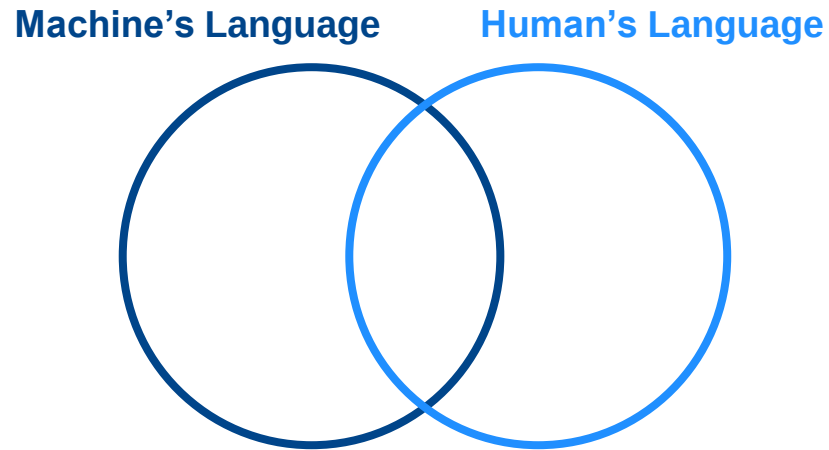
What are concepts?

- “Features are the fundamental units of neural network representations that cannot be further decomposed into simpler independent factors.”
- “A concept is an abstraction of common elements between samples.”
- Minimal, abstract, human understandable...

[Bereska and Gavves - TMLR 2024 - Mechanistic Interpretability for AI Safety — A Review](#)
[Poché, Hervier, and Bakkay – xAI 2023 – Natural Example-Based Explainability: a Survey](#)
[Hewitt, Geirhos, and Kim – ICML 2025 - We Can't Understand AI Using our Existing Vocabulary](#)

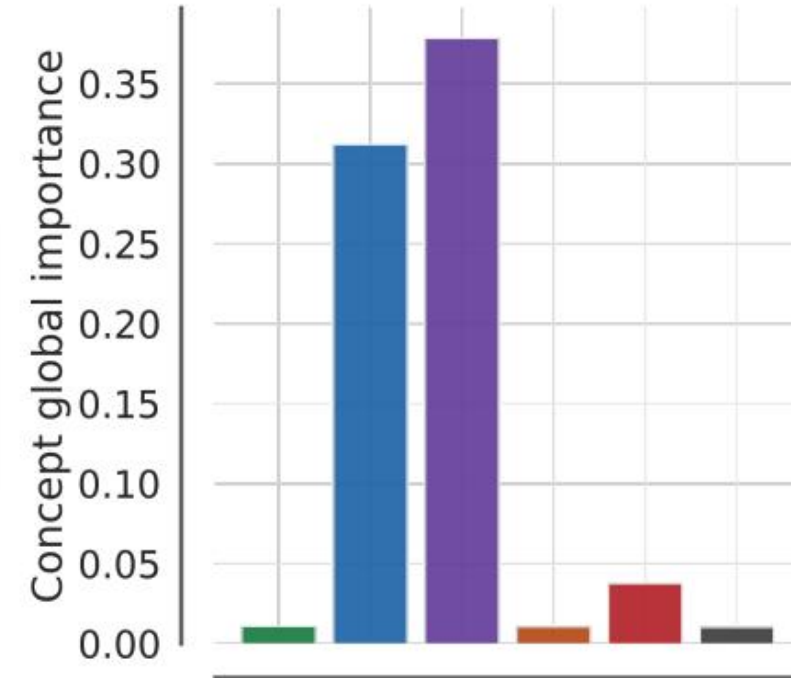
What are concepts?

- “Features are the fundamental units of neural network representations that cannot be further decomposed into simpler independent factors.”
- “A concept is an abstraction of common elements between samples.”
- Minimal, abstract, human understandable...



[Bereska and Gavves - TMLR 2024 - Mechanistic Interpretability for AI Safety — A Review](#)
[Poché, Hervier, and Bakkay – xAI 2023 – Natural Example-Based Explainability: a Survey](#)
[Hewitt, Geirhos, and Kim – ICML 2025 - We Can't Understand AI Using our Existing Vocabulary](#)

What are concept-based explanations?



What are concept-based explanations?

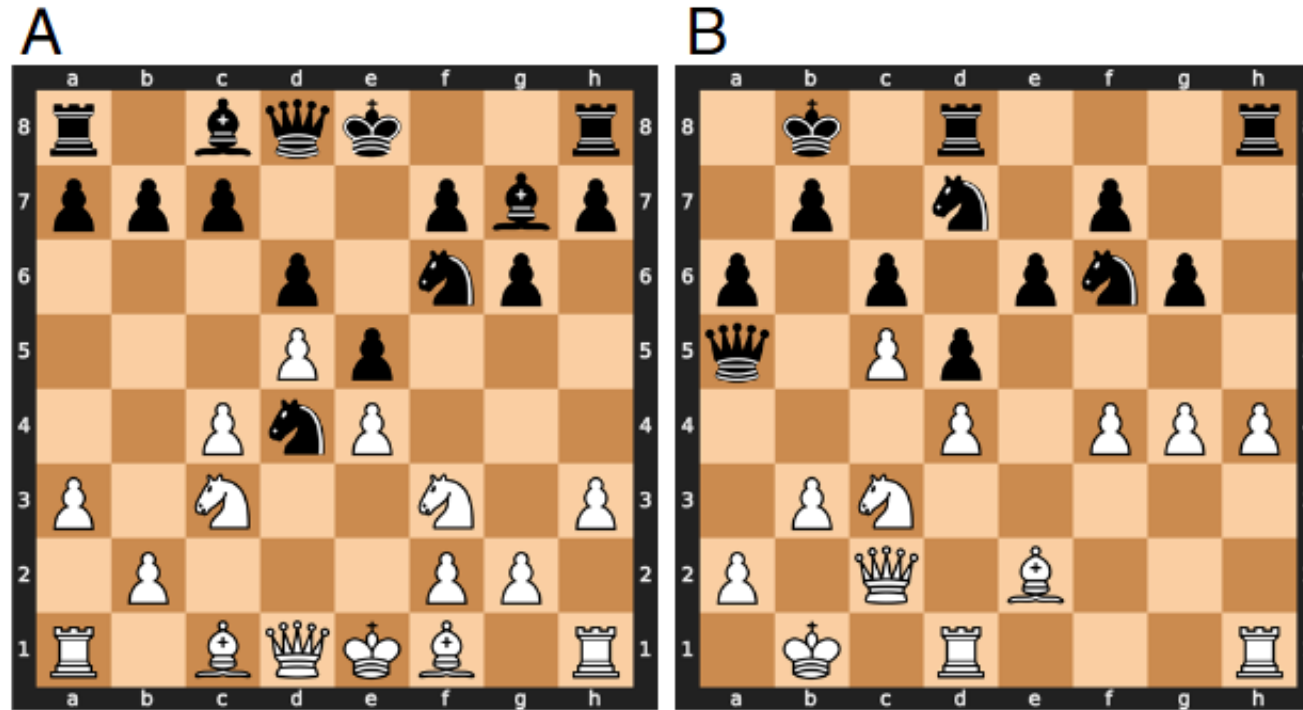
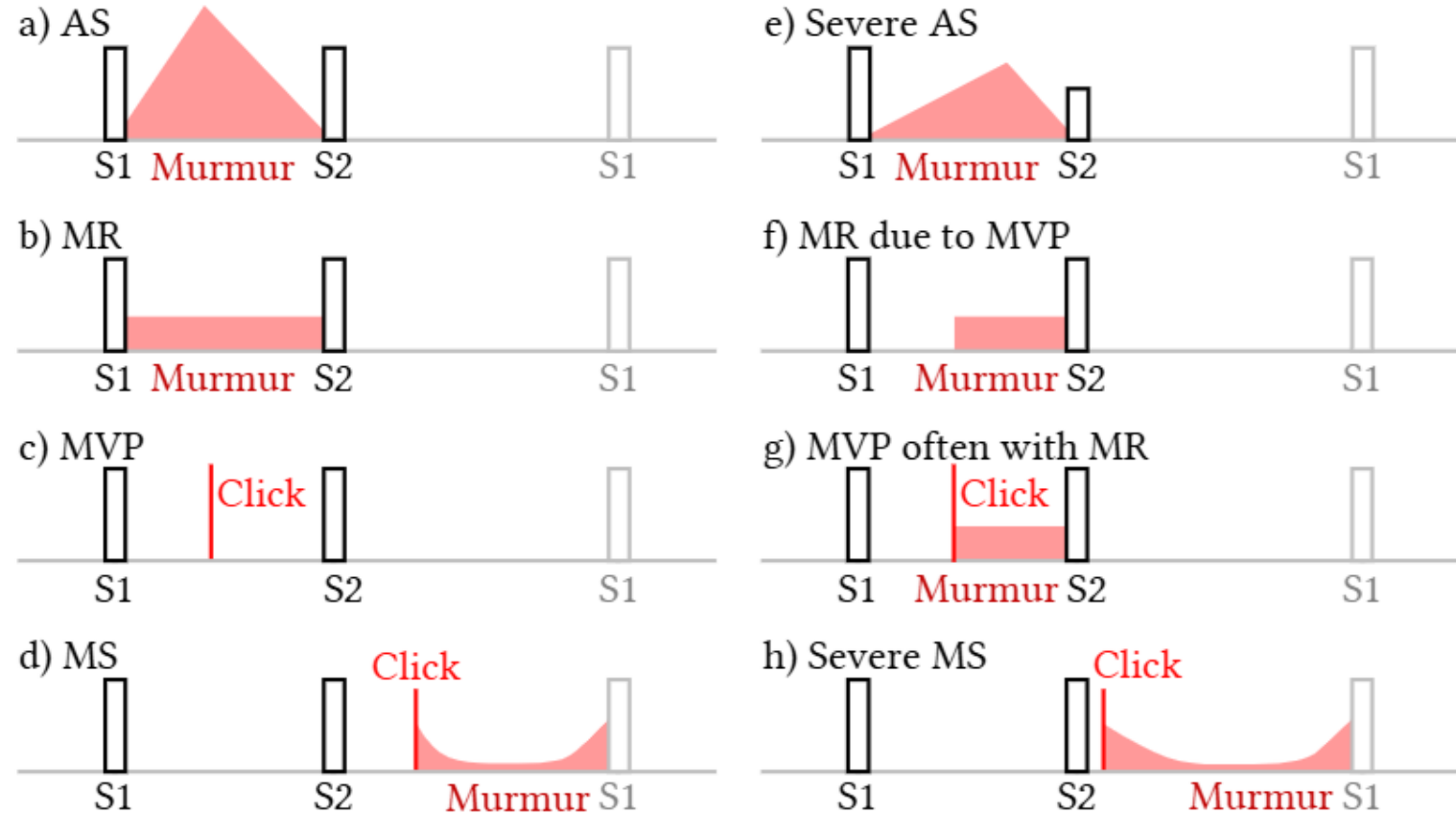
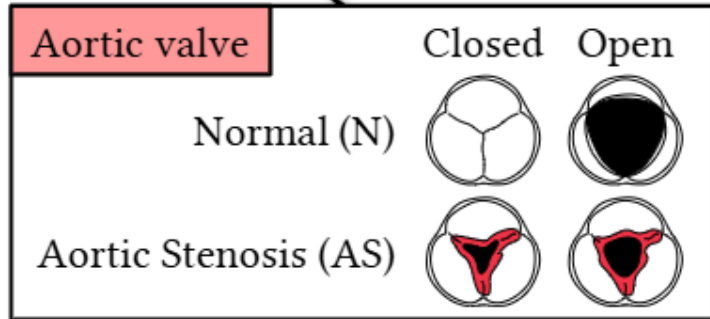
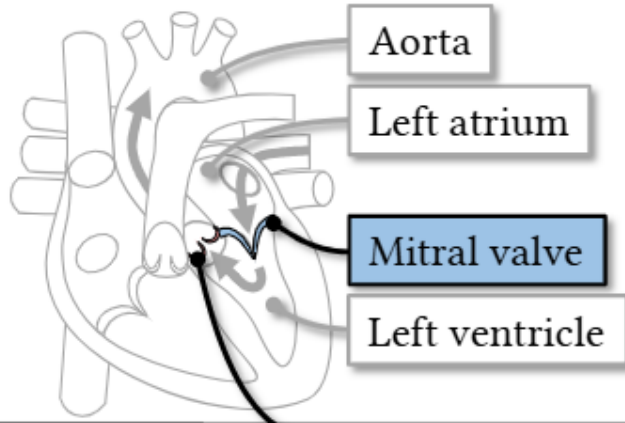


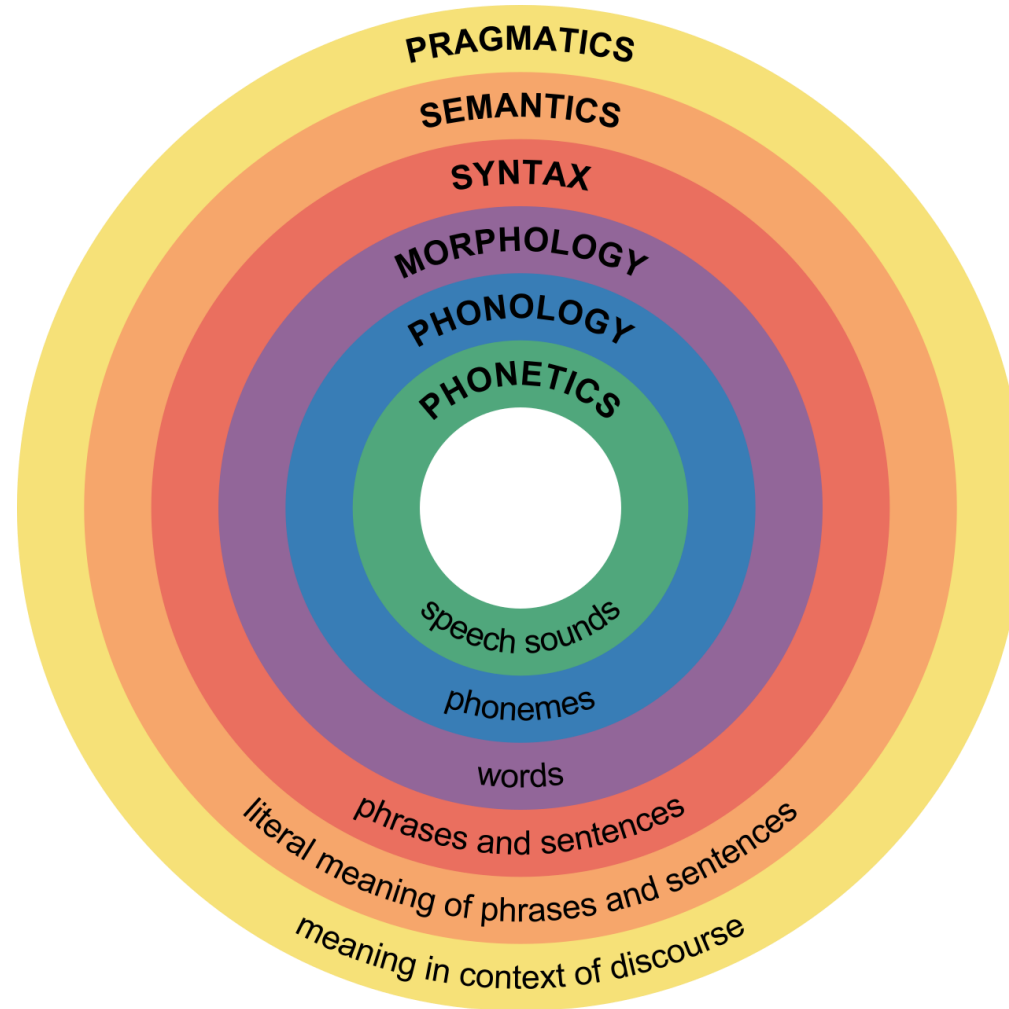
Fig. 6. Concept Puzzles. In both puzzles, White plays a quiet move that improves the piece positioning, and prepares an unconventional long-term sacrifice. (A) In this puzzle, the idea is to play Bg5 to provoke the weakness h6. This weakness allows White to sacrifice the queen in subsequent continuations. (B) In this puzzle, the idea is to play Qd2. In one continuation, White plays b4, an unconventional pawn sacrifice that leads to an advantage.

What are concept-based explanations?

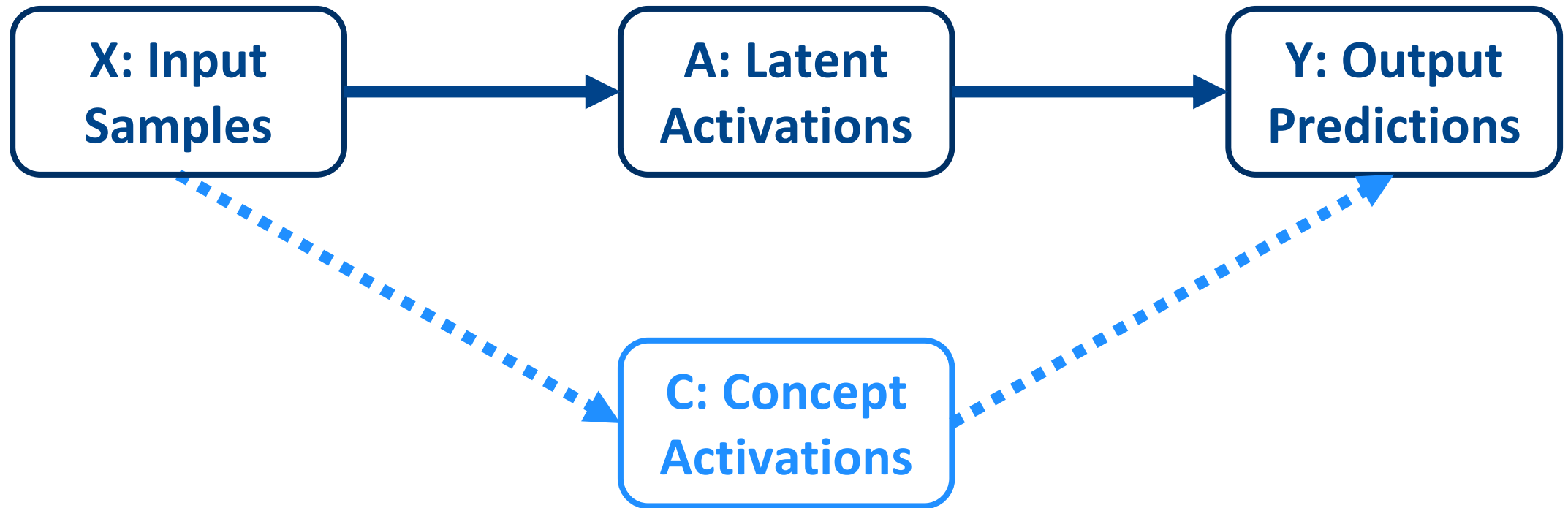
S1 "lub": Mitral valve closing
S2 "dub": Aortic valve closing



What could concepts be in linguistic?



The concept-based explanation diagram



By-design & Supervised

- Concept Bottleneck models



- The model is trained to predict both the concepts and labels with annotations for both.
- Example Loss = $\text{MSE}(C, C^*) + \lambda \text{ BCE}(Y, Y^*)$

By-design & Unsupervised

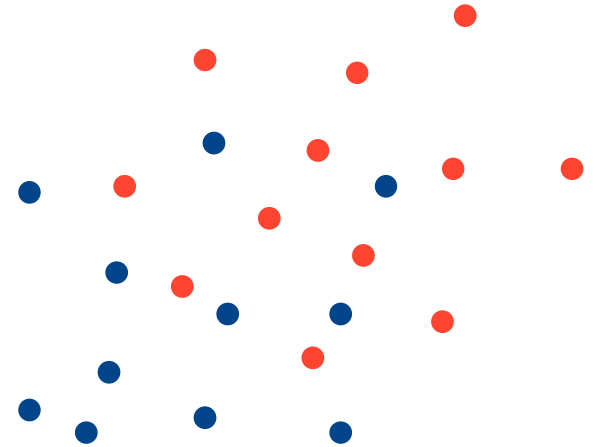
- Prototype-Part-Based Models



- The architecture and/or concept regularization forces properties on concept activations while training to solve the task.
- Example Loss = $L1(C) + \lambda \text{ BCE}(Y, Y^*)$

Post-hoc & Supervised

- CAVs (Concept Activation Vector) and Probes

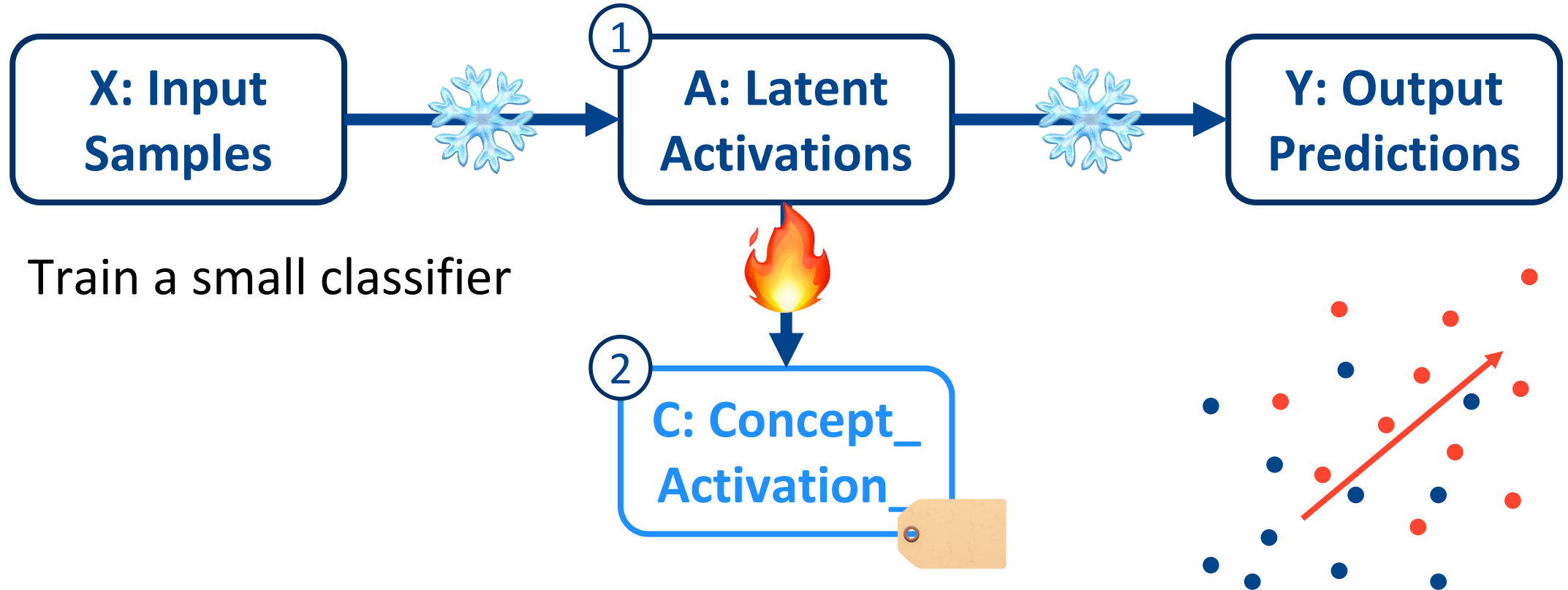


[Alain and Bengio – Stat 2016 - Understanding intermediate layers using linear classifier probes](#)

[Kim et al. - ICML 2018 - Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors \(TCAV\)](#)

Post-hoc & Supervised

- CAVs (Concept Activation Vector) and Probes



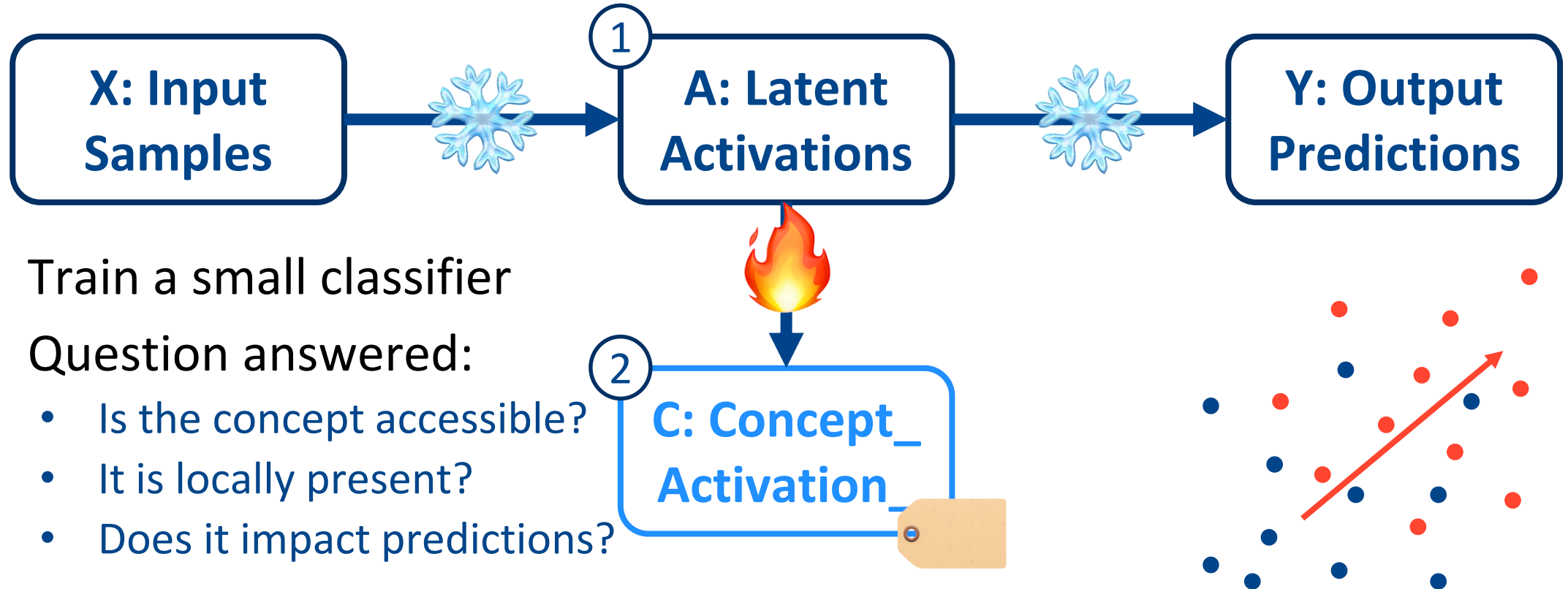
- Train a small classifier

[Alain and Bengio – Stat 2016 - Understanding intermediate layers using linear classifier probes](#)

[Kim et al. - ICML 2018 - Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors \(TCAV\)](#)

Post-hoc & Supervised

- CAVs (Concept Activation Vector) and Probes



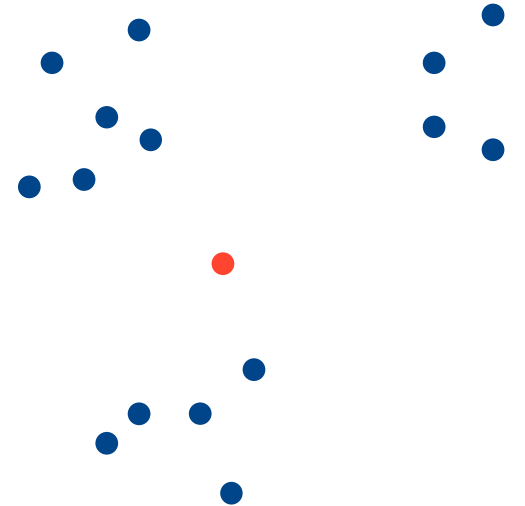
- Train a small classifier
- Question answered:
 - Is the concept accessible?
 - It is locally present?
 - Does it impact predictions?

[Alain and Bengio – Stat 2016 - Understanding intermediate layers using linear classifier probes](#)

[Kim et al. - ICML 2018 - Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors \(TCAV\)](#)

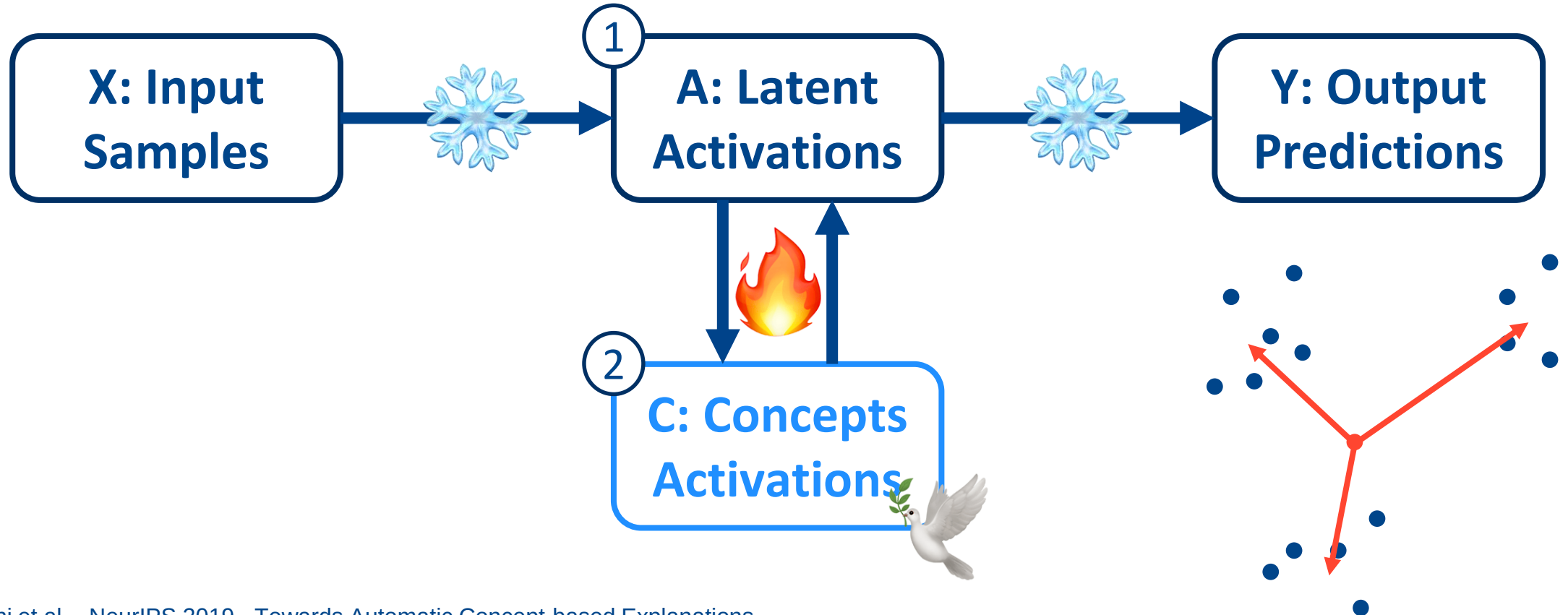
Post-hoc & Unsupervised (SAEs)

1. Split the model and compute a dataset of activations



Post-hoc & Unsupervised

2. Learn concepts mapping (Dictionary learning: $A = CW$)

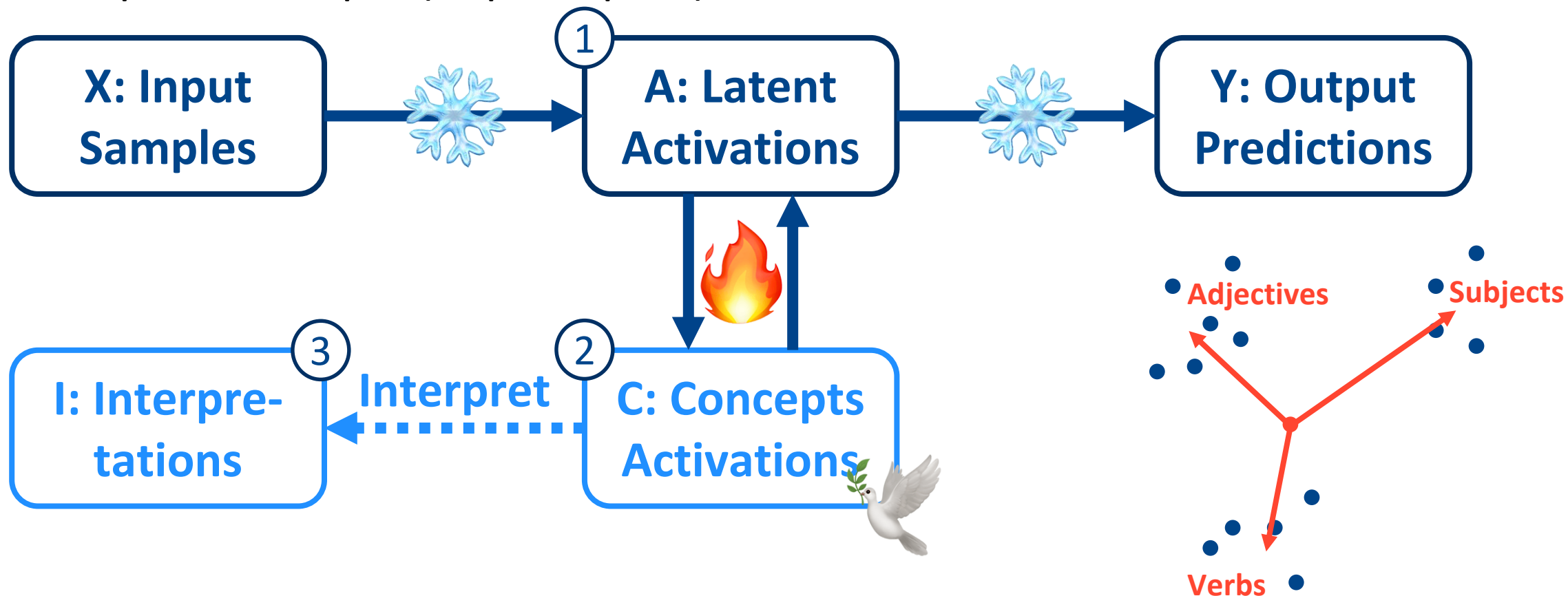


Ghorbani et al. - NeurIPS 2019 - Towards Automatic Concept-based Explanations

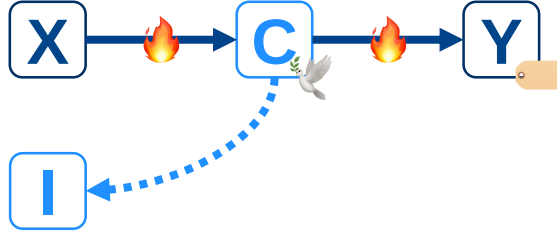
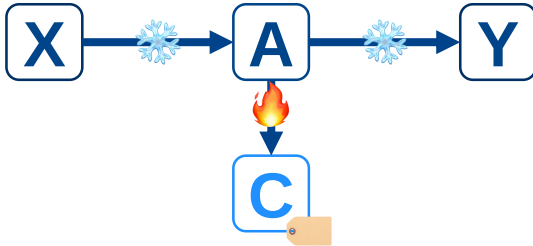
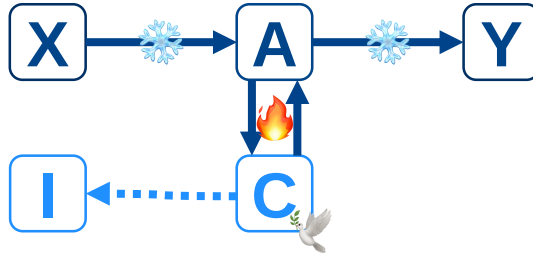
Fel et al. - NeurIPS 2023 - A Holistic Approach to Unifying Automatic Concept Extraction and Concept Importance Estimation

Post-hoc & Unsupervised

3. Interpret concepts (Top-k inputs)



How can we obtain concept-based explanation?

	Supervised 	Unsupervised 	Constraints
By-design 			• Training computation • Loose performance
Post-hoc 			• Require existing model • Finding a A-C mapping
Constraints	• Existing concept labels • Concepts definition through examples	• No guarantee on concept • Concepts need to be interpreted	

Poeta et al. – ACM Computing Surveys - Concept-based Explainable Artificial Intelligence: A Survey

Warnings

- Confirmation bias
- Over-interpretation
- Need for metrics
- Explanations can contradict or complete each others

Tutorials

Applications

Table of content

- Supports:
 - Interpreto
 - BIOS
- Tasks
 - Classification and bias detection
 - Generation
- Explanations formats
 - Attributions
 - Post-hoc concept-based (supervised & unsupervised)
 - Logit Lens?

Interpreto: The Team



Interpreto: An Explainability Library for NLP

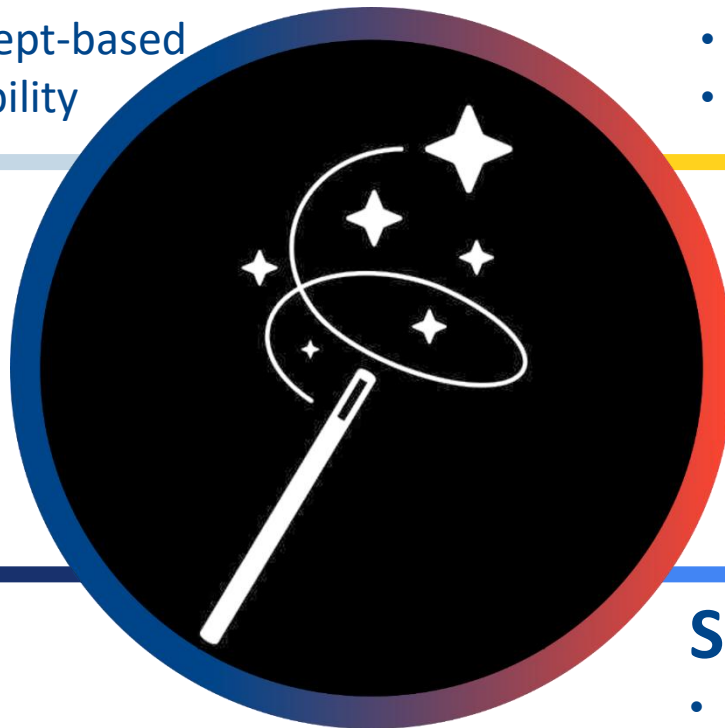
XAI

- Attributions
- Mec Interp / Concept-based
- Post-hoc explainability



NLP

- Any HuggingFace language model
- ForSequenceClassification
- ForCausalLM



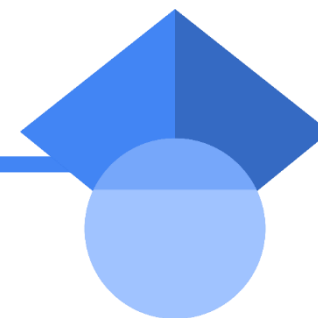
API

- Simple & flexible
- Exhaustive documentation
- Hands-on tutorials



SOTA

- Methods & Metrics
- Reproduce papers from the literature



Interpreto: Differentiation

	Splitting	Learning	Interpretation	Contributions	Metrics	pip & doc
INTERPRETO	✓	✓	✓	✓	✓	✓
NNsight / NNterp / TransformerLens	✓	✗	✗	✗	✗	✓
Overcomplete	●	✓	●	✓	✓	✓
SAELens	✗	✓*	✗	●	✓	✓
Sparsify	✗	✓*	✗	●	✓	●
ViT-Prisma	●	✓*	●	✗	✓	✗
Neuronpedia	✓	✗	✓	✓	✓	●
Delphi	✗	✗	✓	✗	✓	✗
SAE-Vis	✗	✗	✓	✗	✗	✗
SAEBench	✗	✗	✓	✗	✓	●

	Cls	Gen	Metrics	Simple viz	Gran
INTERPRETO	✓	✓	✓	✓	✓
Captum	✓	✓	✓	✗	✗
Ferret	✓	✗	✓	✗	✗
Inseq	✗	✓	✗	✗	✗
SHAP	✓	✓	✗	✓	✗

And post-hoc supervised: Probes

Interpreto: Future Work

- Soon (on going):
 - ViT (vision)
 - Logit Lens
- Later
 - Multimodality (Text + Image) (Speech?)
 - Multi-GPU optimizations

Classification Use Case: BIOS

- Assign an occupation to a biography
- Also includes a binary gender label
- Biased Bios: Remove minority gender samples from occupations

 Fannyjrd/roberta-bios-biased	 Fannyjrd/roberta-bios
 Text Classification •  0.1B • Updated about 22 hours ago •  5	 Text Classification •  0.1B • Updated about 22 hours ago •  2

Model	Dataset	Test Accuracy
RoBERTa-Bios	Bios	86.89 %
RoBERTa-Bios-Biased	Bios	85.39 % (- 1.5%)
RoBERTa-Bios-Biased	Bios Biased	87.79 %

[De-Arteaga et al. - FAccT 2019 - Bias in Bios: A Case Study of Semantic Representation Bias in a High-Stakes Setting](#)

List of tutorials

- Bias detection in classification
 - Application: BIOS
 - Explanations: Attributions, Probes, and unsupervised concepts
- Probing POS in generation models
 - Application: POS (Parts Of Speech)
 - Explanations: Probes
- Attributions of long context generation

